

OXIDATION REDUCTION

Mrinal Kanti Roy

Lecturer (Chemistry)

Mymensingh Engineering College

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- **Oxidizing & Reducing Agent**
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Electronic Concept

According to electronic concept oxidation is defined as

The process in which an atom or an ion loses one or more electrons

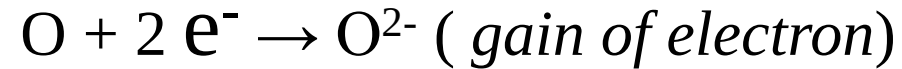
- a. Increase in positive charge
- b. Decrease in negative charge



According to electronic concept reduction is defined as

The process in which an atom or ion gains one or more electrons.

- a. Decrease in positive charge
- b. Increase in negative charge



Classification

1. Redox Reaction

2. Disproportionation reaction

1.Redox Reaction

A reaction in which oxidation and reduction simultaneously take place.

In all redox reactions the total increase in oxidation number must equal the total decrease in oxidation number.

e.g.



2. Disproportionation reaction :

The redox reaction in which same chemical substance / species act as oxidizing as well as reducing agent. Such type of reaction is called disproportionation reaction.

Example :



In above reaction Cl_2 gets reduced into NaCl i.e Oxidation number of Cl_2 decreased from 0 to – 1.

In the same reaction Cl_2 gets oxidized into NaClO i.e oxidation number of Cl_2 increased 0 to 1.

In the above reaction Cl_2 act as both oxidizing and reducing agent. Hence, the reaction is a disproportionation reaction.

OXIDIZING & REDUCING AGENT

- **Oxidizing agent or Oxidant :**

Oxidizing agents are those compound which can oxidize others and reduced itself during the chemical reaction. Those reagents whose O.N. decrease or which gain electrons in a redox reaction are termed as oxidants

e.g. KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, HNO_3 , conc. H_2SO_4 etc. are powerful oxidizing agents.

- **Reducing agent or Reductant :**

Reducing agents are those compound which can reduce others and oxidize itself during the chemical

reaction. Those reagents whose O.N. increase or which loses electrons in a redox reaction are termed as reductants.

e.g. KI , $\text{Na}_2\text{S}_2\text{O}_3$ are powerful reducing agents.

- **Note :**

There are some compounds also which can work both oxidizing agent and reducing agent

OXIDATION NUMBER

At the formation of compounds, the number of electron released to form positive ions and the number of electron received to form negative ion is called the oxidation number of an element.

The oxidation number of a neutral or free element is considered zero (0). When negative ion is formed by accepting electrons, the oxidation number of the element is called negative oxidation number. When positive ion is formed by donating electrons, the oxidation number of the element is called positive oxidation number. Generally the oxidation number of metal is positive, the oxidation number of nonmetal is negative and the oxidation number of radicals follow their charge.

Example: In HCl, oxidation number of H is +1, in H_2 oxidation number of H is 0 (zero), in FeSO_4 oxidation number of Fe is 0 (zero). Similarly in HCl, oxidation number of Cl is -1 and in Cl_2 oxidation number of Cl is 0 (zero).

- Determination of Oxidation number of **Mn** in KMnO_4

Let, oxidation number of Mn = x

Oxidation number of K = +1

Oxidation number of O = -2

As the KMnO_4 neutral compound total oxidation number of atoms is = 0.

So, $(+1) + x + (-2) \times 4 = 0$

$$+1 + x - 8 = 0$$

$$x - 7 = 0$$

$$x = 7$$

Oxidation number of **Mn** in KMnO_4 is = +7

Task: Determine the Oxidation number of the elements of compounds or ions: MnO_2 , $\text{K}_2\text{Cr}_2\text{O}_7$, H_2SO_4 , MnO_4^- , CuSO_4 , NaOH , $\text{Na}_2\text{S}_2\text{O}_3$

Oxidation Number & Valency

1. Oxidation number can be a positive, negative or fractional number but except for inert gases i.e 0, the valency for all element is positive.
2. The oxidation number of the same element differs in the different compound like O.N of carbon in CO , CH_4 , CH_2Cl_2 are +2, +4, +6 respectively. Whereas valency of an element is usually same like valency of carbon in CO , CH_4 , CH_2Cl_2 is 4. However many elements show multiple valencies like valency of copper is 2 or 3.

Difference between Valency and Oxidation Number

Valency

1. Combining capacity of an element.
2. No charge, only a number.
3. It is fixed for an element.
4. It is never zero.
5. Always an whole number.

Oxidation Number

1. Charge which an atom appear after combining.
2. May be positive, negative or fractional value.
3. May differ for an element.
4. Can be zero.
5. May be or may not be a whole number.

Balancing the redox reaction

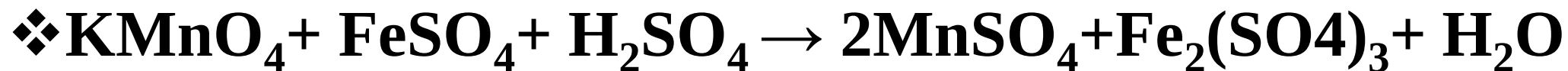
Half reaction method in acidic medium

- Split the reaction into two half – reactions
- Balance elements except H and O
- Balance O by adding H_2O into equations
- Balance H with H^+
- Balance charge by adding electrons as needed
- Multiply each half reaction by an appropriate number to make the number of electrons equal in both half reactions.
- Add two half reactions and simplify where possible by cancelling species appearing in both sides.
- Check equation for same number of atoms and charges on both sides.

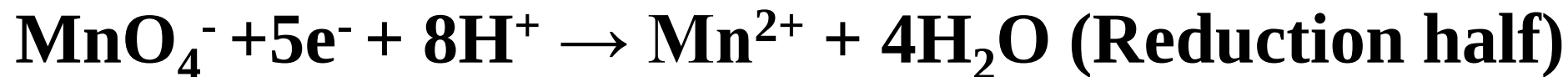
Half reaction method in basic medium

- For balancing redox reactions in basic solutions, all the steps are the same as acidic medium balancing. The H^+ ions can then be neutralized by adding an equal number of OH^- ions to both side of the equations.

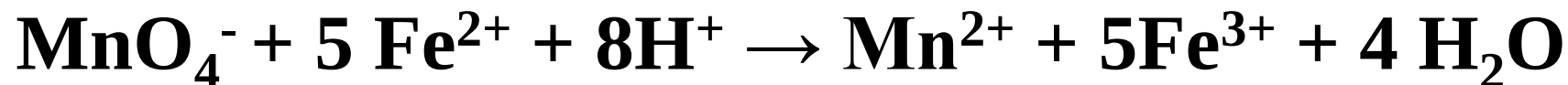
Balance the following redox reaction by ion electron method:



Writing Oxidation and reduction half reaction



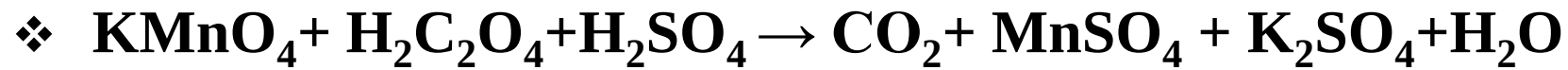
Multiply the oxidation half reaction by 5 and add two half reactions.



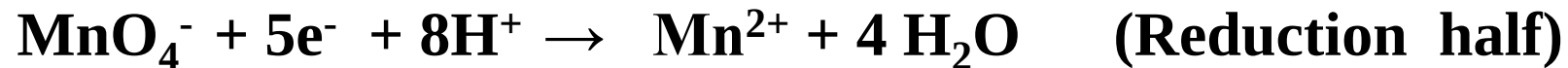
Now balance the ionic reaction into molecular form by adding elements or species or ions.



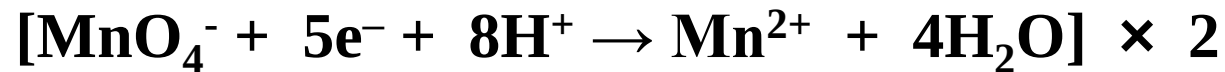
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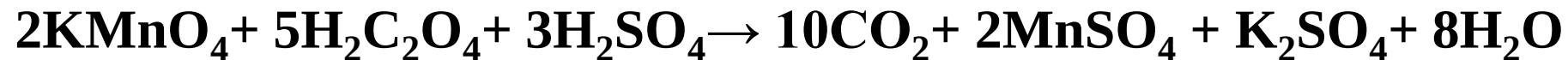
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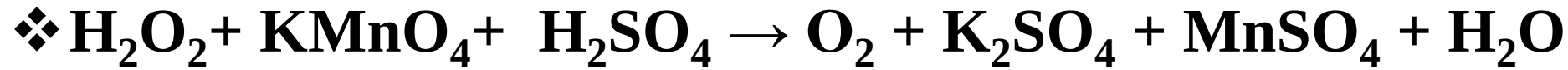
Multiply the oxidation half reaction by 5 and reduction half reaction by 2 to equalize the electrons lost and gained and add the two half reactions



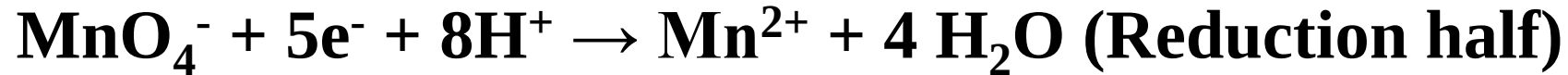
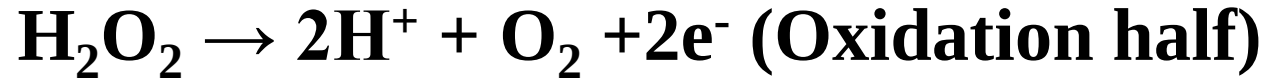
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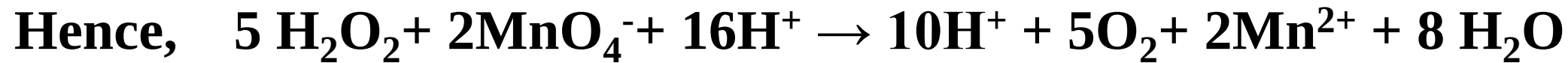
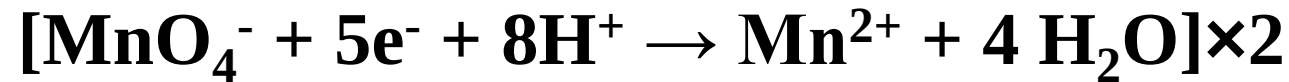
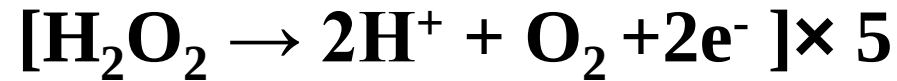
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Writing oxidation and reduction half reaction



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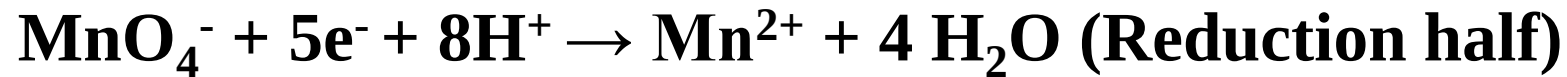
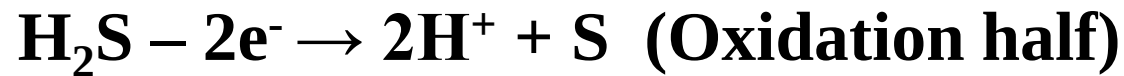
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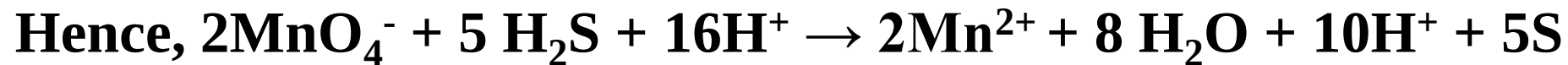
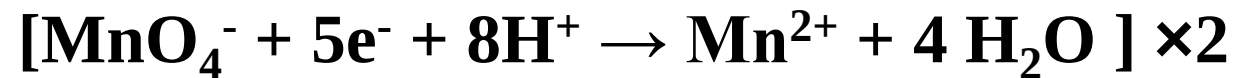
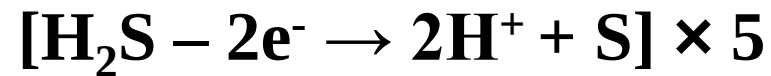
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Writing oxidation and reduction half reaction



Multiply the oxidation half reaction by 5 and reduction half reaction by 2 and add the two half reactions



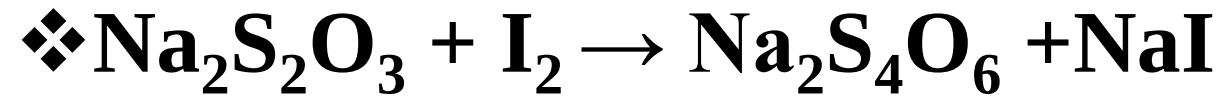
OR



Now balance the ionic reaction into molecular form by adding elements or species or ions.



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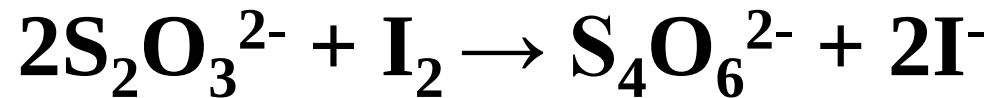
Writing oxidation and reduction half reaction



(Thiosulfate ion) (Tetrathionet ion)



Add the two half reactions



Now balance the ionic reaction into molecular form by adding elements or species or ions.

